FHONEIFY

Pricing Engine Architecture & Evaluation Flow

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This document outlines the step-by-step valuation process implemented by Fhoneify's pricing algorithm to determine fair, market-beating buyback prices against Cashify benchmarks.

STEP 1 Reference Anchor ("Cashify Get Upto")

Every valuation starts with the verified ceiling benchmark for the exact device and storage tier:

```
Fhoneify Public Get Upto = Cashify Reference + Competitor Uplift Bonus
```

This sets the starting reference anchor for a completely flawless, in-warranty device.

STEP 2 Questionnaire Normalization & Brand Routing

User diagnostic responses are filtered and normalized to prevent unfair discrepancies:

- Questions not asked for specific models/age groups are treated neutrally (never penalized as "No").
- Sub-questionnaires (cracks, spots, dents) are validated to ensure consistency.
- The device request is dispatched to specialized brand models (Apple, Samsung, Xiaomi, OnePlus, Vivo, etc.).

STEP 3 Basic Condition, Age & Warranty Deductions

Evaluates the phone's fundamental operational viability and legal status:

- **Calling Connectivity:** Failure to make/receive calls triggers an immediate drop to the scrap value floor (e.g. ₹1,180 for iPhone 18 Pro Max, ₹200–₹1,200 for Android).
- **Touch & Screen Authenticity:** Non-working touch or aftermarket replacement screen incurs major baseline deductions.
- **Age / Warranty / GST Invoice:** Out-of-warranty devices (>11 months) or devices without a valid GST bill receive age-bracketed depreciation.

STEP 4 Functional & Hardware Fault Deductions

Specific component tests are mapped to discrete repair/refurbishment deductions:

- **Motherboard / IC:** Severe penalties for Wi-Fi or Bluetooth controller failures.
- **Biometrics:** Face ID failure incurs substantial penalties; Touch ID is penalized only on devices containing fingerprint sensors.
- **Battery Health:** Calibrated penalties for "Service Required" (<80%) vs. degradation (80%–85%).
- **Peripherals:** Cameras, speaker, earpiece, charging port, microphone, and physical buttons.

STEP 5 Physical & Cosmetic Defect Deductions

Cosmetic wear and structural damage are classified across dedicated matrices:

- **Display Glass & Matrix:** Glass crack vs. outer chipping vs. scratches; heavy spots vs. light spots; visible lines; discoloration.
- **Body & Frame:** Minor/major dents, body scratches, bent/curved frame, or loose screen gap.
- **Overlap Damping:** Simultaneous screen and body wear applies an overlap damping factor to avoid excessive double-penalization.

STEP 6 Retail Packaging & Accessories

Adjustments for original packaging and accessories:

- **Original Box with IMEI:** Retains full value (absence incurs ~₹900–₹2,250 deduction).
- **Original Charger:** Evaluated based on whether the original manufacturer shipped an adapter in-box.

STEP 7 Cashify Condition Equivalent Calculation

Calculates the net depreciated price reflecting Cashify's exact buyback quote for that condition:

```
Cashify Condition Equivalent = Base Reference - Total Validated Deductions
```

STEP 8 Fhoneify Competitor Uplift (Tiered Bonus)

Fhoneify applies a competitive bonus to guarantee higher customer payout than market competitors:

Budget (≤ ₹20k)	Mid-Tier (₹20k - ₹50k)	Flagship (> ₹50k)
+8%	**+6%**	**+4%**

```
Fhoneify Final Price = Cashify Equivalent + min(Reference * Rate, ₹1,000 Cap)
```

STEP 9 Floor Protection & Guaranteed Inspection Quote

- **Floor Price:** Final offer is guaranteed never to drop below the safety floor (₹100).
- **Fee Transparency:** Accounting for validation and inspection fees guarantees the exact net cash disbursed to the customer's bank account or UPI.